ARTICLE

Systemically delivered lipid nanoparticle-mRNA encoding lysosomal acid β-glucosidase restores the enzyme deficiency in a murine Gaucher disease model

Yuanqing Liu[1]✉, Shasha Wang[1], Yanni Chen[1], Zhang Zhang[2], Xiaojiang Quan[1], Zhijun Guo[2] and Zihao Wang[2]

© The Author(s), under exclusive licence to Springer Nature Limited 2025

Gaucher disease (GD) is a rare genetically inherited illness caused by loss of lysosomal acid β-glucosidase (β-GCase) that leads to progressive accumulation of substrates, sphingolipid glucosylceramide (GL1) and glucosylsphingosine (lyso-GL1). The protein-based enzyme replacement therapy (ERT) requires frequent dosing due to short drug half-life causing challenges in long-term patient compliance. JCXH-301 is a lipid nanoparticle (LNP) encapsulated messenger RNA (mRNA) encoding β-GCase. Intravenous administration of JCXH-301 delivered the target mRNA to various tissues in mice with intracellular expression of β-GCase predominantly in macrophages and dendritic cells in the spleen and bone marrow. In *GBA1* D427V homozygous mice treated with JCXH-301, the dose-dependent in vivo production of functional β-GCase resulted in reduction of serum lyso-GL1, a key biomarker of GD. The therapeutic effect of JCXH-301 was sustained for a duration significantly longer than that of protein-based ERT Cerezyme. JCXH-301 administration induced minimal pro-inflammatory cytokines in the liver and spleen. Taken together, these results provide proof-of-concept for using LNP-delivered mRNA as a new drug modality to restore the β-GCase genetic deficiency for GD treatment.

Gene Therapy (2025) 32:599–609; https://doi.org/10.1038/s41434-025-00549-2

INTRODUCTION

GD is a lysosomal storage disorder caused by mutations in the glucosylceramidase beta 1 (*GBA1*) gene that encodes β-GCase. More than 400 mutations spanning along the *GBA1* gene have been identified in relation to GD [1, 2]. These mutations result in β-GCase deficiency leading to accumulation of glucosylceramide (GL1) and its deacylated form glucosylsphingosine (lyso-GL1) [3, 4]. β-GCase protein presents most conspicuously in lysosomes of cells of the reticuloendothelial system (RES), the eponymous Gaucher cells [5], which locate mainly in the liver, spleen, bone marrow, but also lymph nodes and other organs [6]. As GD may affect multiple systems, symptoms in patients vary widely. Type 1 GD mainly displays systemic symptoms, including hepatosplenomegaly, anemia, leukopenia, and bone abnormalities [2]. The type 2 and 3 are neuronopathic forms of GD characterized by astrogliosis, neuronophagia, and neuronal loss [3, 4]. The type 1 GD is the most prevalent form in Caucasian descendants accounting for over 90% of cases, while in other populations, neuronopathic forms are more prevalent [7]. The accumulation of GL1 and lyso-GL1 is implicated in mediating immune dysregulation and skeletal disease. Plasma lyso-GL1 is currently considered to be the most sensitive and predictive GD-specific biomarker [8, 9].

There are two types of specific treatment for GD, ERT and substrate reduction therapy (SRT). ERT supplies exogenous β-GCase thereby slowing down the course of disease in GD patients, while SRT decreases the production of GL1 by inhibiting the glucosylceramide synthase. Both treatments are effective in treating type 1 GD but improve little the neurologic outcomes of type 2 and 3 GD patients and are expensive [10, 11]. Several ERT drugs are commercially available, including the recombinant human Imiglucerase (Cerezyme®, Sanofi, Cambridge, USA). These protein-based drugs are unexceptionally short-lived after dosing, for example, a half-life of 3.6 to 10.4 min for Cerezyme (product monograph of Cerezyme, https://pdf.hres.ca/dpdpm/ 00077549.PDF). Thus, developing a novel ERT modality with longer half-life and lower cost remains an unmet medical need.

Advances in mRNA chemistry have greatly improved the safety profile of non-immunostimulatory mRNA-based therapies [12–14]. In the context of GD, direct delivery of mRNA to cells via LNP offers a considerable advantage over the conventional ERT therapy because the coding mRNA can utilize the intracellular machinery to produce the target protein at desired cellular localization for therapeutic benefit [11, 15, 16]. Compared to DNA gene delivery, mRNA therapy corrects protein disfunctions without modifying the genomic DNA [17, 18]. Furthermore, the transient protein expression by mRNA mitigates the risk of unintentional overdose due to constitutive protein function. In this report, a synthetic mRNA encoding murine β-GCase was encapsulated with LNP to form an LNP-mRNA complex, designated as JCXH-301m. Intravenous (i.v.) administration of JCXH-301m delivered the *GBA1* mRNA to multiple organs, with cells of RES being remarkably targeted. Examining β-GCase in the liver and spleen (main onset organs of GD), and in serum, revealed that the duration of β-GCase

[1]Immorna (Shanghai) Biotechnology, Co. Ltd., Shanghai, China. [2]Immorna (Hangzhou) Biotechnology, Co. Ltd., Hangzhou, Zhejiang, China. ✉email: yuanqing.liu@immorna.com

Received: 9 December 2024 Revised: 29 May 2025 Accepted: 25 June 2025
Published online: 3 July 2025

expression by JCXH-301m was remarkably longer than Cerezyme administration. The increased β-GCase enzymatic activity led to a significant reduction of lyso-GL1 both in the circulation and in the disease inflicted organs after JCXH-301m administration. Repeated JCXH-301m administration induced only minimal and transient innate cytokine production, indicating good tolerability. Finally, JCXH-301 was shown to express the functional human β-GCase in vivo. These results highlight the potential of developing JCXH-301 as a novel ERT treatment modality to restore the β-GCase genetic deficiency in GD patients.

MATERIALS AND METHODS

Animals

C57BL/6 mice of 6–8 weeks age (Shanghai Lingchang Biotechnology Co., Ltd., Shanghai, China) were housed in Shanghai Jiaotong University. The *GBA1* D427V homozygous transgenic mice (C57BL/6JSmoc-*Gba*$^{em1(D427V)Smoc}$ aging between 20 to 64 weeks), harboring an A to T transversion at the 119th nucleotide of exon 9 of *GBA1* gene, were purchased from and housed in Shanghai Model Organisms Center, Inc. All mice were housed under pathogen-free conditions in isolated ventilated-cages. The mice were maintained on a 12/12-h light/dark cycle at 20–26 °C with sterile pellet food and water *ad libitum*. For experiments using D427V mice, animals were bred per request and genotyped to confirm the D427V homozygosity. Male and female D427V homozygous mice were then allocated to the experimental groups with effort made to match the age and sex of animals per time point/group. No randomization was used. In each experiment, 12–21 animals per group, 3 animals per time point were used for kinetical analysis of treatment effects. The investigator was blinded to the group allocation during the experiment but was not blinded when assessing the outcome. All experiments were independently repeated at least once with similar results obtained.

mRNA synthesis and LNP formulation

In vitro transcription of mRNA with N1-Methylpseudo-UTP modification was carried out at 30 °C for 3 h with 300 rpm vertex using a linearized DNA template containing gene of interest (all reagents from Hongene Biotech, Shanghai, China). The mRNA product was capped at 37 °C for 90 min using Vaccinia virus capping enzyme and 2′-O-methyltransferase (Hongene Biotech) after removing DNA template by Turbo DNase (Thermo Fisher Scientific, Waltham, MA, USA). The mature mRNA was purified by LiCl (Thermo Fisher Scientific) precipitation.

The mRNA encapsulation was carried out by fast mixing of aqueous phase and ethanol phase using a microfluidic device (INano™ L system, Micro & Nano, Shanghai, China). For aqueous phase, mRNA was dissolved in 50 mM citrate buffer (pH 6.0). The ethanol phase comprised XH-08, cholesterol, 1,2-diastearoyl-sn-glycero-3-phosphocholine (DSPC), and 1,2-dimyristoyl-rac-glycero-3-methoxypolyethyleneglycol-2000 (DMG-PEG2000). XH-08 is an ionizable cationic lipid with an alcohol functionality in the headgroup. Comparing to the dimethylamine headgroup, LNPs formed with XH-08 has a slightly higher surface pKa around 6.56 ± 0.13, which may enhance the chemical stability of the LNPs and be important for the in vivo performance of LNP-mRNA products. Four lipid components were mixed at molar ratios of 40:48:10:2 (XH-08: cholesterol: DSPC: DMG-PEG2000). After subsequential dialyzing, concentrating and filtrating, the final formulation was characterized for particle size, polydispersity index, mRNA concentration and encapsulation efficiency respectively.

Cell transfection

The BHK cells were purchased from the National Collection of Authenticated Cell Cultures (UK) and routinely tested for mycoplasma contamination. BHK cells were seeded into six-well plates at 3×10^5 cells/well and cultured overnight before transfection with mRNA using Lipofectamine 3000 (Thermo Fisher Scientific). The RIPA lysis buffer (Beyotime, Shanghai, China) was used to lyse cells. The supernatant of cell lysate was collected and stored at −80 °C for further analysis.

Protein collection

Samples of serum, liver and spleen were collected from euthanized mice. Liver and spleen samples were frozen immediately after collection and stored at −80 °C for further process. Subsequently, tissues were homogenized in RIPA lysis buffer with a tissue breaker at 60 Hz for 180 s at 4 °C). The supernatant was collected after centrifuging at 16,000 × *g* for 30 min at 4 °C.

Western blot

Boiled protein samples of 40 μg were loaded in SDS-polyacrylamide gel for electrophoresis, then transferred to a nitrocellulose (NC) membrane (General Electric, Boston, USA). After blocking with 5% non-fat milk in PBST, the membrane was incubated with primary and respective secondary antibodies. Specific protein bands were visualized using a Bio-RAD ChemiDoc™ MP imaging system (Bio-Rad, California, USA). Mouse anti-β-GCase (Merck ABN1387, Darmstadt, Germany; 1:1000), rabbit anti-β-actin (Beyotime AF5003, Shanghai, China; 1:5000) or rabbit anti-GAPDH (Proteintech 60004-1-PBS, Wuhan, China; 1:2000) antibodies were used respectively.

Detection of β-GCase enzymatic activity

The enzymatic activity of β-GCase was measured as previously described [19]. Briefly, the tissue lysate or serum was mixed with substrate at 37 °C for 60 min together with 1 M Glycine at pH 10.5. The 4-methylumbelliferon product was measured at excision 365 nm/emission 445 nm using a microplate reader (Thermo Fisher Scientific). The specific activity unit was determined as pmol/min/μg.

Quantification of lyso-GL1 by UPLC-MS/MS

Serum lyso-GL1 extraction was based on a modified method from Bligh and Dyer [20]. Briefly, 50 μL of 80% methanol was added to 5 μL serum with 500 pg of N, N-Dimethyl-D-*erythro*-spingosine (Matreya, Pennsylvania, USA) as an internal standard [9]. The sample mixture was vortexed for 3 min followed by centrifuging (16,000 × *g* at 4 °C for 10 min). The supernatant was collected for UPLC-MS/MS analysis. The lyso-GL1 standard (Matreya) was prepared similarly to described above for a calibration curve. For tissue samples, 0.1 g wet tissue was homogenized in 1 mL 80% methanol followed by centrifuging as above. The supernatant was collected for UPLC-MS/MS.

The UPLC-MS/MS analysis was conducted on an UPLC system equipped with a Q-Exactive orbitrap mass spectrometer (Thermo Fisher Scientific). Separation of components was achieved using an Acquity BEH C18 column (2.1 × 100 mm, 1.7 μm column, Waters) at 50 °C. Mobile phase A (0.1% formic acid in water) and B (0.1% formic acid in acetonitrile) were operated with a gradient elution. Electrospray ionization (ESI) was in positive mode. The scan data collection was executed in parallel reaction monitoring mode with the following transitions: m/z 462.34253 > 282.27901/264.26848 for lyso-GL1 and m/z 328.32101 > 310.30960/280.29920 for N, N-Dimethyl-D-*erythro*-sphingosine, with a resolution of 70,000. Both system and data collected were processed by Chromeleon™ software (Thermo Fisher Scientific).

Assessment of *GBA1*, cytokine mRNA level in tissues by qPCR

Total RNA was extracted with RNAiso plus (Takara, Shiga, Japan) and reverse transcribed to cDNA using a PrimeScript RT reagent Kit with gDNA Eraser (Takara) according to manufacturer's instruction. The qPCR reaction was performed with ChamQ Universal SYBR qPCR Master Mix (Vazyme, Nanjing, China) using primer sets specific to *GBA1*, or to the respective cytokine genes on the QuantStudio™ 6 Flex Real-Time PCR system (Thermo Fisher Scientific). The fold change to saline was calculated according to formular: $2^{\wedge -(\Delta Ct\ treated\ -\ \Delta Ct\ control)}$.

Analysis of GFP positive cell subsets by flow cytometry

Spleen and bone marrow were collected from naïve C57BL/6 mice at 4 or 24 h after i.v. administration of LNP encapsulated GFP-mRNA. Subsequently, single-cell suspensions were prepared by mashing spleen or bone marrow in 1 mL PBS through 70-μm cell strainer, followed by lysis of red blood cells. Cells were then stained with Live/Dead Zombie Violet cell dye (Biolegend, San Diego, USA), followed by TruStain FcX PLUS (Biolegend 156603) to block Fc receptors. Cells were subsequently incubated with a mixture of fluorescent dye-conjugated antibodies against CD45-PerCP/Cy5.5 (Biolegend 103132), CD3-APC/Cy7 (Biolegend 100222), CD19-PE (Biolegend 152408), CD49b-PE/Dazzle 594 (Biolegend 108924), F4/80-PE/Cy7 (Biolegend 123114), CD11b-Brilliant Violet 510 (Biolegend 101263), MHCII-Brilliant Violet 650 (Biolegend 107641) and CD11c-Brilliant Violet 785 (Biolegend 117336) to label immune cell subsets. The GFP expression in immune cell subsets were measured by CytoFLEX (Beckman Coulter, Indianapolis, USA) and analyzed with CytExpert software (version 2.4).

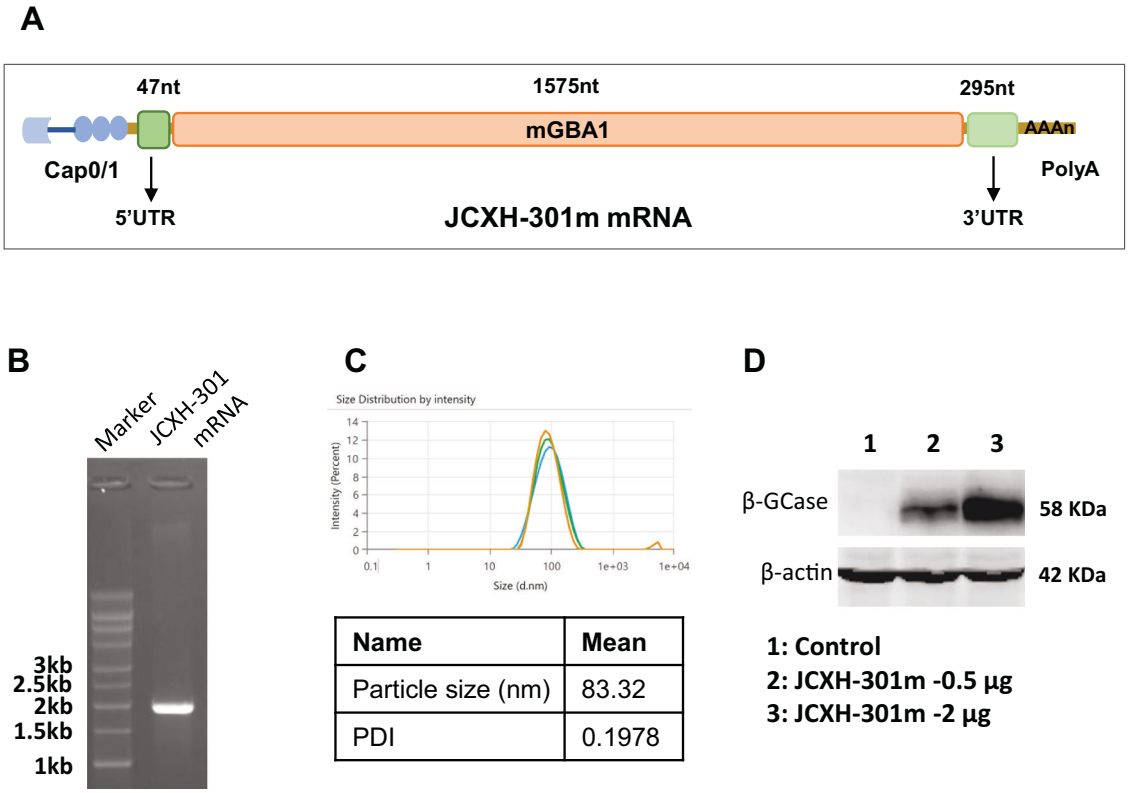


Fig. 1 Structure and characterization of JCXH-301m. A Schematic diagram of the JCXH-301m mRNA structure consisting of Cap0/1, 5'UTR, mouse GBA1 ORF, 3'UTR and PolyA tail. **B** Agarose gel shows the JCXH-301m mRNA band of 2015 nucleotides (nt) after in vitro transcription and capping. **C** The size and PDI of JCXH-301m after LNP encapsulation. **D** Expression of mouse β-GCase (58 kDa) in BHK cells 24 h after transfection of JCXH-301m by Western blot.

Statistical analysis

All statistical tests were selected based on the experimental design and are appropriate for the data structure. Normality was assessed using the Shapiro-Wilk test on the residuals of each data set. Homogeneity of variances was tested using Bartlett test or Brown-Forsythe test. After parametric analysis passed these assumptions ($p > 0.05$), one-way ANOVA followed by Dunnett's multiple comparisons test was performed to compare multiple treatment groups to a single control; or two-way ANOVA followed by Sidak's multiple comparisons test was used, where two independent factors (e.g., treatment × time) were assessed simultaneously. Statistical analyses were performed using GraphPad Prism 8 software.

RESULTS

Design, production and characterization of JCXH-301m

An mRNA encoding murine β-GCase was synthesized and encapsulated with LNP to form an LNP-mRNA complex, designated as JCXH-301m. The mRNA of JCXH-301m contains a Cap0/1, 5' UTR, mouse *GBA1* sequence (ID: 14466), 3' UTR and poly A tail (Fig. 1A). The uridine in the mRNA was replaced by N1-Methylpseudo-UTP during IVT.

It was noticed that three currently approved ERT products are recombinant β-GCase with a few differences in the amino-acid sequence. To select a suitable sequence for mRNA drug, three mouse *GBA1* sequences that resembled those ERT products were compared for the expression of β-GCase in vitro. A Taliglucerase alfa-like *GBA1* (Tali-L) contained R495H, EF at N-terminal and ELLVDTM at C-terminal; an Imiglucerase-like *GBA1* (Imi-L) had R495H; and a Velaglucerase alfa-like *GBA1* (Vela-L) was the natural mouse *GBA1* sequence. BHK cells were transfected with the mRNA encoding each *GBA1* using lipo3000. As shown in supplementary Fig. 1, the β-GCase expression of three mRNAs was similar at both 24 and 48 h. Hence, the nature *GBA1* sequence was chosen for JCXH-301m in further experiments.

JCXH-301m mRNA was encapsulated in LNP comprised of ionizable cationic lipid XH-08, cholesterol, DMG-PEG2000 and phospholipid DSPC. Characterization of JCXH-301m showed a single band in agarose gel indicating high mRNA quality (Fig. 1B). JCXH-301m showed greater than 90% encapsulation efficiency, a particle size of 80–100 nm and polydispersity index (PDI) of less than 0.3 (Fig. 1C). Transfection with JCXH-301m in BHK cells resulted in the expression of β-GCase in a dose dependent manner (Fig. 1D).

Distribution of JCXH-301m in naïve mice after i.v. administration

The distribution of JCXH-301m mRNA was evaluated after a single i.v. injection of JCXH-301m (0.5 mg/kg) in naïve mice. Compared to non-treated animals, the *GBA1* mRNA level was significantly increased in the liver, spleen, lung, and bone marrow within 12 to 24 h in the JCXH-301m-treated animals (Fig. 2). A trend of *GBA1* mRNA increase was also observed in the brain and kidney. The fold-increase was 11996 ± 1986, 608 ± 149, 142 ± 72, 92 ± 12, 80 ± 15, 5.4 ± 0.6 respectively in the lung, bone marrow, kidney, spleen, liver, and brain at 12 h (Fig. 2A–F).

We next examined the main cellular targets of JCXH-301m in the spleen and bone marrow. A LNP encapsulated mRNA encoding GFP was i.v. injected to naïve mice like JCXH-301m treatment. The GFP+ cells among total splenocytes or bone marrow cells were examined by flow cytometry 4 and 24 h later (Fig. 2G). Normal spleens contained around 20–30% of T and B lymphocytes each, and low levels of natural killer (NK) cells, macrophages, and dendritic cells (DC). In the bone marrow, B lymphocytes were the most abundant while other immune cells were at lower levels. The GFP expression was found predominantly in the macrophage population in both spleen (ranging

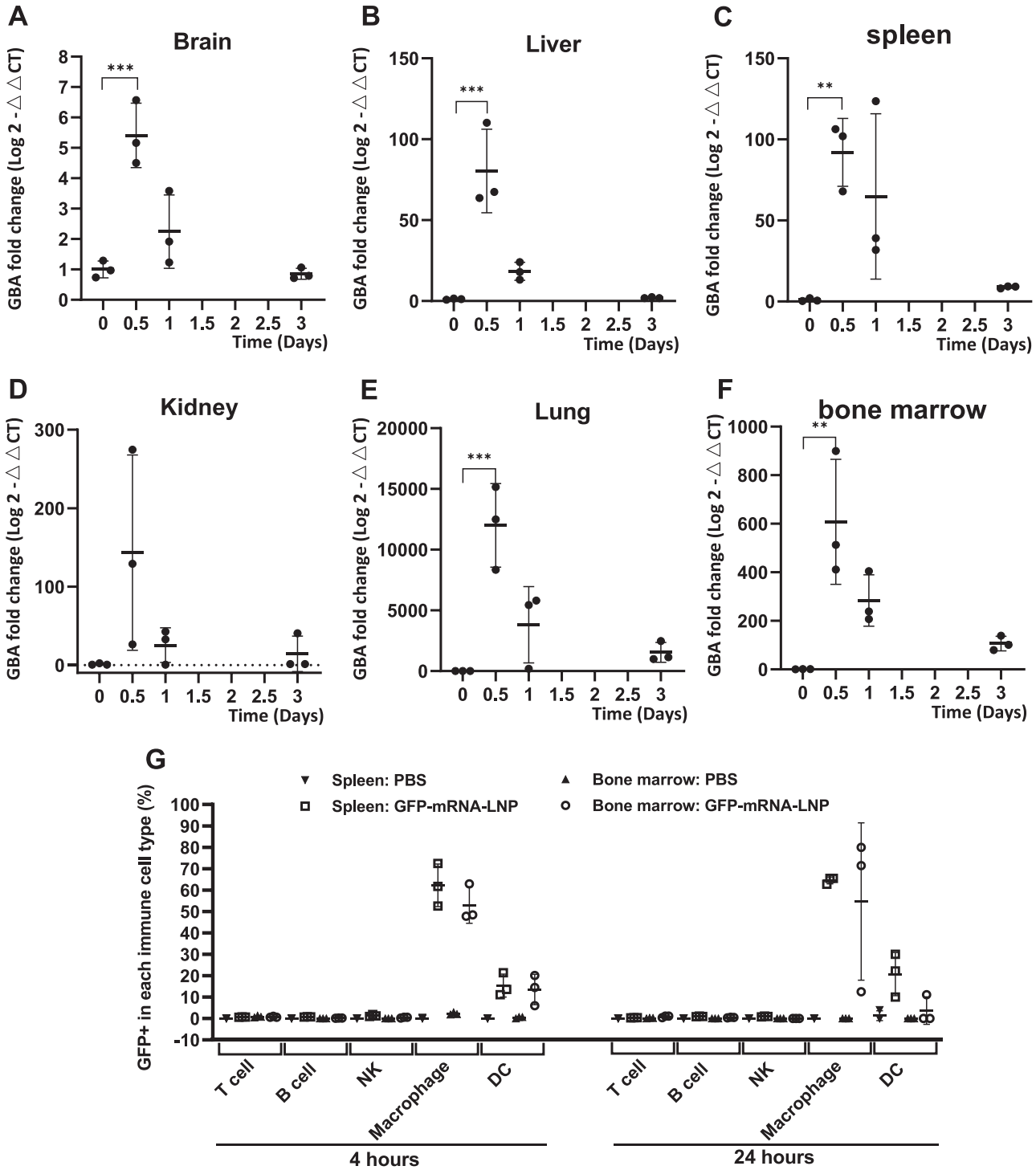


Fig. 2 Distribution of JCXH-301m after i.v. administration in naïve mice. A–F Naïve mice were treated once intravenously with JCXH-301m (0.5 mg/kg). Tissues of brain (**A**), liver (**B**), spleen (**C**), kidney (**D**), lung (**E**), and bone marrow (**F**) were harvested at indicated time points after treatment. The level of *GBA1* mRNA was determined by measuring fold changes of *GBA1* mRNA in copy numbers of JCXH-301m treated animals relative to untreated controls using qPCR. Fold change was calculated as follows: $Log_2(\Delta\Delta Ct) = Log_2$ [(Ct *GBA1* – Ct *GAPDH*) $_{\text{treated JCXH-301m}}$/(Ct *GBA1* – Ct *GAPDH*) $_{\text{control}}$]. The differences between values at day 0 and day 0.5, 1 or 3 were analyzed statistically using one-way ANOVA followed by Dunnett's multiple comparisons test ($n = 3$, ***$P < 0.0005$, **$P < 0.005$). **G** Expression of GFP in immune cell subsets in the mouse spleen and bone marrow were analyzed by flow cytometry at 4 h or 24 h after i.v. injection of LNP encapsulated GFP-mRNA ($n = 3$). Dot plots displayed the proportion of GFP positive cells in each immune cell subset in spleen and bone marrow. T cells (CD45+ CD3+), B cells (CD45+ CD19+), NK cells (CD45+ CD3- CD19- CD49b+), macrophages (CD45+ CD3- CD19- CD49b- CD11b+ / low F4/80), and dendritic cells (CD45+ CD3- CD19- CD49b- F4/80- MHCII+ CD11c+) were measured. Each symbol represented one mouse. This experiment was repeated once with similar result obtained.

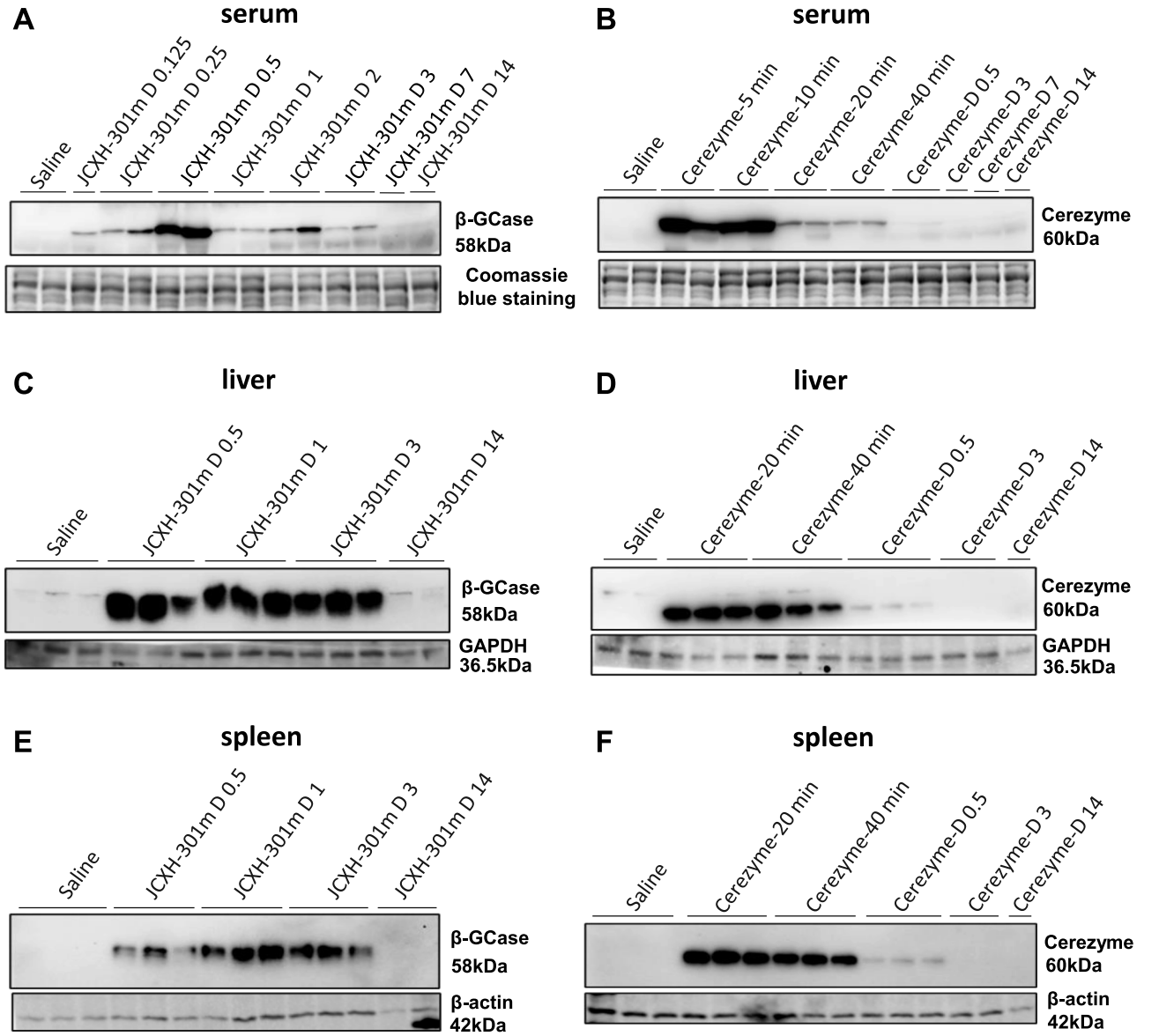


Fig. 3 Expression of β-GCase in *GBA1* D427V mice after i.v. administration of JCXH-301m or Cerezyme. *GBA1* D427V mice were treated once intravenously with 0.5 mg/kg JCXH-301m, 60 U/kg Cerezyme or PBS, respectively. The level of β-GCase protein was detected in serum (**A**, **B**), liver (**C**, **D**) and spleen (**E**, **F**) by western blot. Protein loading quantity was controlled using Coomassie blue staining for serum, GAPDH blotting for liver, or β-actin blotting for spleen. The short solid line on top of gel picture indicates the parallel individual samples. This experiment was repeated at least once with similar result obtained.

62.2–53.0%) and bone marrow (ranging 64.6–54.6%) as soon as 4 h after drug administration. There were 15.4% and 20.7% of splenic DC cells, 13.6% and 3.7% of DC cells in bone marrow that also expressed GFP at 4 and 24 h after treatment. However, minimal or no GFP signal was detected in T, B, and NK cells in both organs (Fig. 2G).

Kinetics of β-GCase expression in *GBA1* D427V mutant mice after JCXH-301m therapy

The *GBA1* D427V mutant mouse is a canonical GD animal model [21]. These animals exhibit a reduced β-GCase enzymatic activity and associated accumulation of lyso-GL1 in target tissues [22, 23]. Therefore, it is pertinent to evaluate the in vivo pharmacokinetics and pharmacodynamics of JCXH-301m in *GBA1* D427V mice. As a comparison, Cerezyme was used on a dose previously used in mice [24].

Firstly, the β-GCase expression was evaluated in serum, liver, and spleen. No or marginal serum β-GCase protein was detected in saline-treated *GBA1* D427V mice as expected. Following i.v. administration of JCXH-301m (0.5 mg/kg), the level of serum β-GCase increased at 3 h and peaked at 12 h. The expression remained detectable up to 3 days after treatment (Fig. 3A). However, the serum β-GCase level peaked at 5–10 min after i.v. administration of Cerezyme but rapidly decreased at 20 min (Fig. 3B). Similarly, elevation of β-GCase persisted much longer in the liver and spleen of animals treated by JCXH-301m than by Cerezyme (Fig. 3C–F). In particular, high levels of β-GCase were detected from 12 h to at least 3 days in liver and spleen of JCXH-301m treated *GBA1* D427V mutant mice (Fig. 3C, E). On a contrary, the elevated β-GCase reached peak at 20 and 40 min following Cerezyme treatment but deceased to a barely detectable level by 12 h in the liver and spleen (Fig. 3D, F).

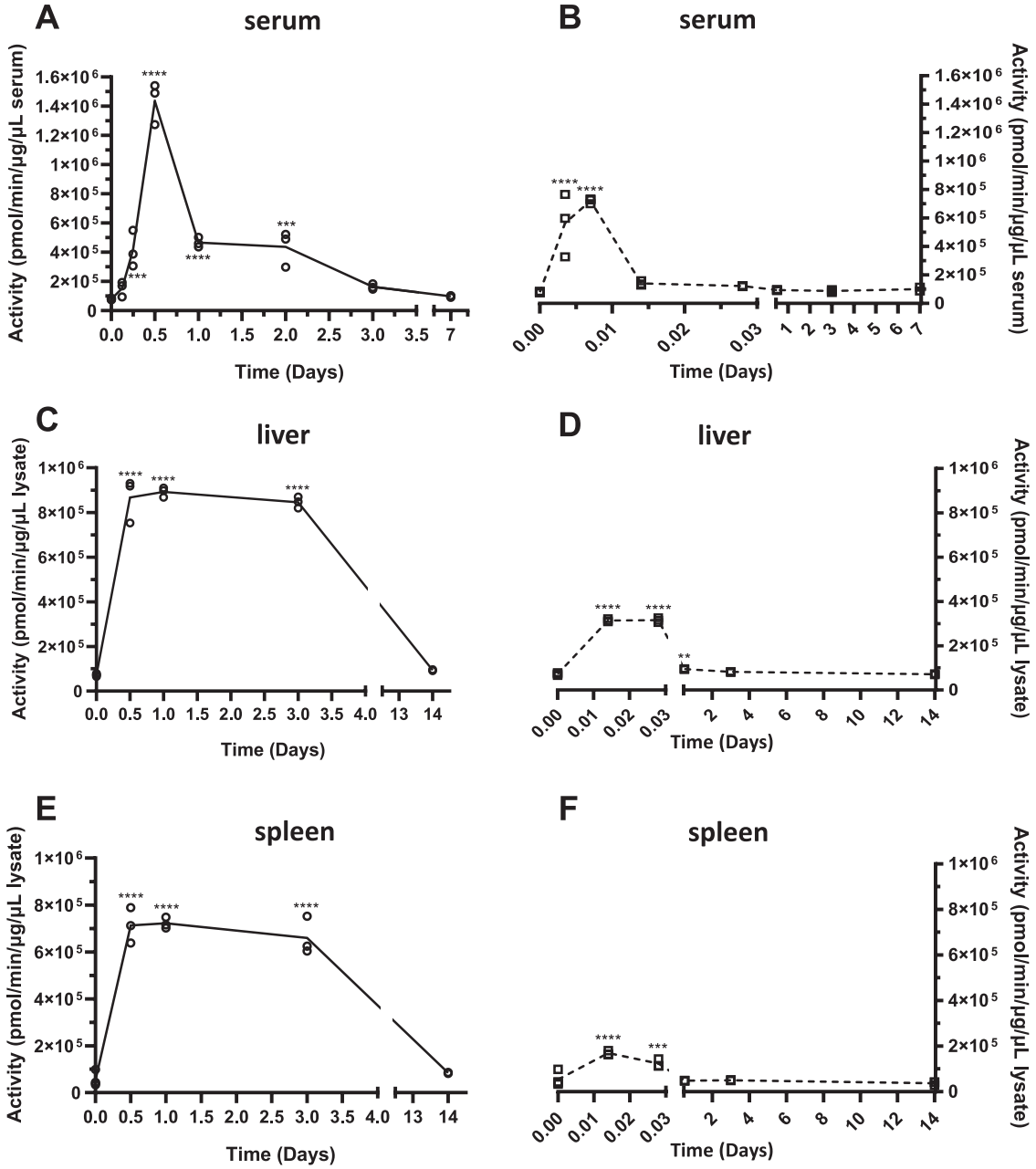
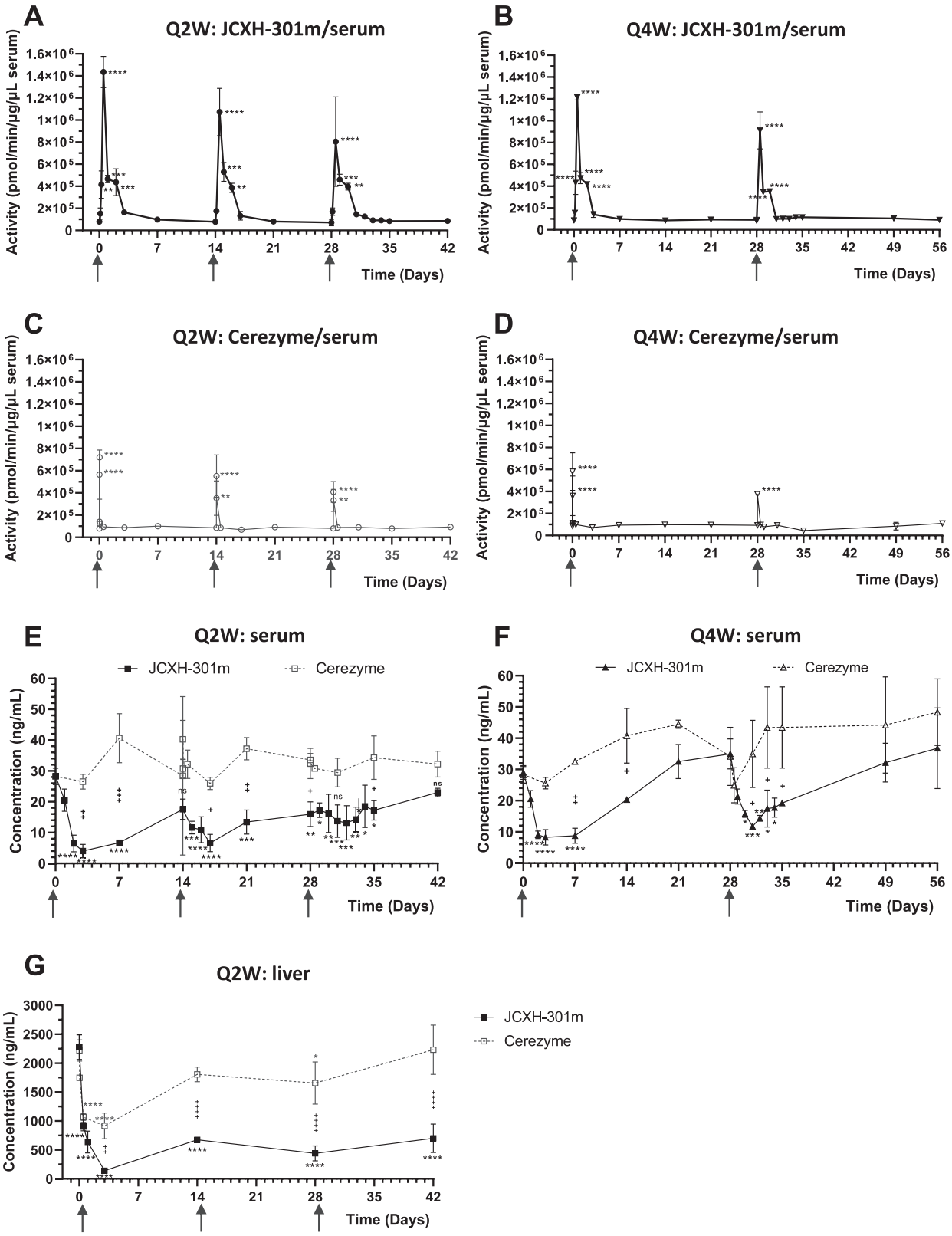


Fig. 4 Enzymatic activity of β-GCase in *GBA1* D427V mutant mice after i.v. administration of JCXH-301m or Cerezyme. *GBA1* D427V mice were treated once with 0.5 mg/kg JCXH-301m, 60 U/kg Cerezyme or PBS, respectively. The enzymatic activity of β-GCase was measured in serum (**A**, **B**), liver (**C**, **D**) and spleen (**E**, **F**). The solid line showed the enzymatic activity with treatment of JCXH-301m, and the dashed line showed the enzymatic activity after Cerezyme injection. Asterisks indicate a significant difference between the treatment group and vehicle group using one-way ANOVA followed by Dunnett's multiple comparisons test ($n = 3$, ****$P < 0.0001$, ***$P < 0.0005$, **$P < 0.005$). This experiment was repeated at least once with similar result obtained.

Enzymatic activity of β-GCase in serum and target organs after JCXH-301m therapy

In addition to protein expression level, the β-GCase enzymatic activity in the serum and tissue samples was measured to further assess the effectiveness of JCXH-301m therapy in ameliorating the defective β-GCase function. An increased β-GCase activity was recorded post i.v. administration of either JCXH-301m or Cerezyme, but with totally different kinetics. The JCXH-301m therapy resulted in a rise of enzymatic activity that peaked (17-fold increase) at 12 h in the serum (1.4×10^6 vs. 8×10^4 pmol/min/μg/μL, $P < 0.001$). At 7 days after treatment, it decreased near to baseline though still significantly higher than the pre-administration level (Fig. 4A). On a drastic contrary, the β-GCase activity increased quickly and reached peak (9-fold of the pre-dose level) at 10 min after Cerezyme administration (7.2×10^5 vs. 8.0×10^4 pmol/min/μg/μL, $P < 0.001$), followed by a rapid decrease to baseline at 12 h, in consistent with previously reported data [24].

After JCXH-301m treatment, the β-GCase enzymatic activity peaked on day 1 in liver (13-fold increase, 9×10^5 vs. 7×10^4 pmol/min/μg/μL, $P < 0.001$) and in spleen (14-fold increase, 7×10^5 vs. 5×10^4 pmol/min/μg/μL, $P < 0.001$). The enzymatic activity maintained at a high level for at least 3 days before returning to baseline on day 14 (Fig. 4C, E). Nevertheless, the β-GCase activity peaked shortly after Cerezyme administration at 40 min in liver (4-fold increase, 3×10^5 vs. 7×10^4 pmol/min/μg/μL, $P < 0.001$) and at 20 min



in spleen (3-fold increase, 1.7×10^5 vs. 5×10^4 pmol/min/μg/μL, $P < 0.001$) before returning to the baseline at 12 h in both liver and spleen (Fig. 4D, F). These data confirmed that JCXH-301mtreatment led to a prolonged functional β-GCase activity in the serum, liver, and spleen comparing to Cerezyme treatment.

Efficacy of JCXH-301m in *GBA1* D427V mutant mice after repeat-dose i.v. administration

GD is a life-long disease requiring repeated drug administration. To mimic a possible clinical treatment regimen, the *GBA1* D427V mutant mice were given multiple doses of JCXH-301m (0.5 mg/kg)

Fig. 5 Effect of multiple injections of JCXH-301m or Cerezyme in *GBA1* D427V mutant mice. *GBA1* D427V mice were i.v. administrated with 0.5 mg/kg JCXH-301m, 60 U/kg Cerezyme or PBS every other week for three times (Q2W: **A**, **C**, **E**, **G**) or every 4 weeks for two times (Q4W: **B**, **D**, **F**). Arrows indicate the time points of injections. **A**, **B** The enzymatic activity of β-GCase in serum after administration of JCXH-301m. **C**, **D** The enzymatic activity of β-GCase in serum after administration of Cerezyme. The concentration of lyso-GL1 after administration of JCXH-301m or Cerezyme in serum (**E**, **F**) and in liver (**G**). Error bars represent standard errors on arithmetic means. The difference between the timepoints after administration versus the pre-dose level were analyzed statistically using one-way ANOVA followed by Dunnett's multiple comparisons test. Asterisks indicate a significant difference ($n = 3$, ****$P < 0.0001$, ***$P < 0.0005$, **$P < 0.005$). The difference between JCXH-301m and Cerezyme at the same timepoint was analyzed statistically using two-way ANOVA followed by Sidak's multiple comparisons test. Cross signs indicate a significant difference ($n = 3$, $^{++++}P < 0.0001$, $^{+++}P < 0.0005$, $^{++}P < 0.005$, $^{+}P < 0.05$). This experiment was repeated once with similar result obtained.

by i.v. injection, either every two weeks (Q2W) for 3 times or every four weeks (Q4W) for 2 times (indicated by arrow, showed in Fig. 5A–F). Consistent with observations after a single dose administration, JCXH-301m treated *GBA1* D427V mice showed a statistically significant increase in serum β-GCase activity when compared to pre-dose from 3 h till 3 days with a peak at 12 h after each drug administration (Fig. 5A, B). More interestingly, the enhancement of β-GCase activity was sustained at the same magnitude without significant difference after each JCXH-301m injection over the course of repeat-dose treatment, although the peak of β-GCase activity appeared to be lower at the subsequent doses than the first dose (Fig. 5A, B). Enhancement of β-GCase activity was also observed in the Cerezyme treated animals, however persisted for a much shorter duration as compared to JCXH-301m administration, lasting only for 40 min (Fig. 5C, D).

The serum lyso-GL1 level is a key biomarker of GD. The reduction of lyso-GL1 is an indicate for therapeutic effect. Therefore, the concentration of lyso-GL1 was measured to verify the eventual therapeutic effect in JCXH-301m treated animals. Like in GD patients, the concentration of lyso-GL1 was elevated in the serum of *GBA1* D427V mice. Lyso-GL1 started to decrease as soon as 3 h after either JCXH-301m or Cerezyme treatment, then reached the bottom after 3 days (Fig. 5E, F). However, the fold of reduction was 10 times greater with JCXH-301m (10-fold reduction to baseline) than with Cerezyme (0.9-fold reduction to baseline) after the first injection. Moreover, the reduced lyso-GL1 level in serum was maintained for 2 weeks in the JCXH-301m treated animals (Fig. 5E, F), whereas serum lyso-GL1 level returned rapidly to the pre-dose level on Day 7 after Cerezyme treatment (Fig. 5E, F). Consequently, by JCXH-301m repeated dosing, particularly for Q2W, the drug treated animals maintained a serum lyso-GL1 level persistently below the baseline (Fig. 5E). Whereby the serum concentration of lyso-GL1 returned to baseline in 1–3 days after a transient decrease upon Cerezyme administration (Fig. 5E, F).

Strikingly, in the liver of *GBA1* D427V mice, the lyso-GL1 level rapidly decreased from a mean value of 2276 ng/mL to 140 ng/mL at 72 h after JCXH-301m administration. Moreover, lyso-GL1 was maintained below the baseline level at a mean value between 400 ng/mL to 700 ng/mL in liver upon Q2W repeat dosing (Fig. 5G). In contrast, repeated Cerezyme administration every two weeks was insufficient to maintain a reduced lyso-GL1 level in the liver despite of early lyso-GL1 reduction to a mean of 915 ng/mL at 12 h after the first injection (Fig. 5G), then gradually returned to a mean value of 1800 ng/mL on day 14, and maintained at that level even after the following 2 injections (Fig. 5G). Meanwhile, stimulation of innate cytokines could indicate the tolerability of JCXH-301m therapy. Livers and spleens were collected from D427V mutant mice during JCXH-301m treatment (Q2W). The levels of interferon alpha (IFN-α), interferon beta (IFN-β), interferon gamma (IFN-ɣ), interleukin-6 (IL-6), tumor necrosis factor alpha (TNF-α), and retinoic acid-inducible gene I (RIG-I) were measured by qPCR, and the fold-change relative to the levels in the saline-treated control mice was determined. As shown in supplementary Fig. 2A, C, a slight, transient but significant elevation of RIG I, IL-6 and TNF-α in the liver at 12 h; IFN-α and IL-6 in the spleen at 12 h; RIG I in the spleen at 12 h and 1 day was noted after the first JCXH-301m administration. No significant increase for these proinflammatory mediators was noted in liver or spleen at other examined timepoints after repeated JCXH-301m treatment. It is known that concanavalin A (Con A) triggers acute liver inflammation [25, 26]. As a positive control, the level of innate cytokines in livers and spleens of C57BL/6 mice was examined by qPCR at 8 and 24 h after i.v. injection of Con A (3 mg/kg). Indeed, the Con A treatment induced drastic and more sustained elevation of IFN-β and IFN-ɣ in the liver (supplementary Fig. 2B), and IFN-β and IFN-α in the spleen (supplementary Fig. 2D). Putting together, these results demonstrate the absence of persistent innate immune stimulation by JCXH-301m therapy.

Expression of functional human β-GCase by JCXH-301 in vivo in *GBA1* D427V mice

Finally, we constructed an LNP encapsulated mRNA encoding human β-GCase (Gene ID: 2629), named JCXH-301, using the same process as JCXH-301m. After i.v. injection of JCXH-301 in *GBA1* D427 mice. The expression of β-GCase peaked at Day 1 in both liver and spleen (Fig. 6A, B) with an apparent dose effect (except for the low-dose group in the spleen, which peaked on day 3). The β-GCase level remained above the pre-dose level up to 7, 3 and 1 day in the liver after JCXH-301 administration at high, middle and low dose, respectively, while the expression level of β-GCase in the spleen was higher in the low and medium dose groups than in the high dose group at day 3 after dosing. JCXH-301 administration resulted in a rise of β-GCase enzymatic activity in the serum, liver and spleen (Fig. 6C–E). The effect was dose-dependent reaching a 5-fold, 2.9-fold, or 2.0-fold increase to peak at 12 h in the serum (Fig. 6C) by 0.5, 0.1 or 0.02 mg/kg of JCXH-301 dosing respectively as compared to the pre-administration level ($P < 0.0001$ for 0.5 and 0.1 mg/kg group; $P < 0.05$ for 0.02 mg/kg group). Similarly, after JCXH-301 administration at 0.5, 0.1 or 0.02 mg/kg, the β-GCase enzymatic activity peaked at day 1 in liver (9.0-fold, 7.3-fold, or 3.7-fold increase respectively to the pre-dose level, $P < 0.0001$ for all) (Fig. 6D) and in spleen (6.7-fold, 5.4-fold or 2.7-fold increase respectively to the pre-dose level, $P < 0.0001$ for 0.5 and 0.1 mg/kg group; $P < 0.05$ for 0.02 mg/kg group) (Fig. 6E). The elevated enzymatic activity was maintained for 3 days in serum and even longer in liver and spleen before returning to pre-dosing level. These data confirmed that JCXH-301 therapy led to prolonged functional β-GCase persistence in serum, liver, and spleen of *GBA1* D427V mice in a dose-dependent manner.

DISCUSSION

GD is the most common lysosomal storage disorder and the first to be successfully treated by ERT [27]. Until now, three protein-based β-GCase have been approved for the ERT treatment of GD. However, frequent administration and exceedingly high cost of current ERT have led to continuous searching for longer lasting and more cost-effective therapeutic modalities [28, 29]. Gene

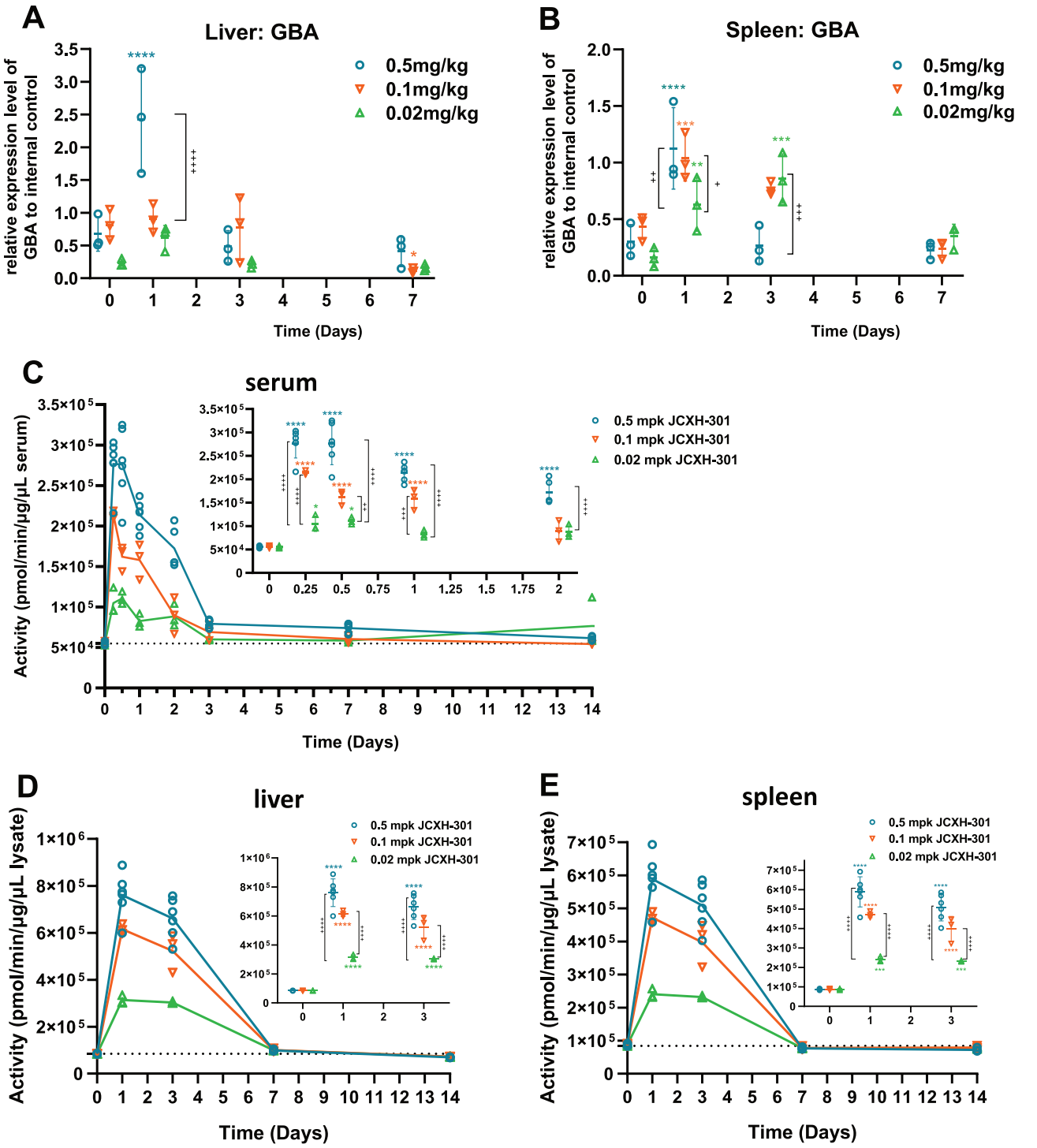


Fig. 6 The human β-GCase protein level and enzymatic activity in *GBA1* D427V mutant mice after i.v. injection of JCXH-301. The β-GCase protein level and activity after i.v. injection of 0.5 mg/kg, 0.1 mg/kg, or 0.02 mg/kg JCXH-301 in *GBA1* D427V mice. The relative β-GCase protein level detected by western blot in (**A**) liver and (**B**) spleen. The enzymatic activity of β-GCase in serum (**C**), liver (**D**) and spleen (**E**). The inset graphs in (**C**–**E**) show the extended peak periods with statistical analysis. Error bars represent standard errors on arithmetic means. The difference between the treatment group and vehicle group was analyzed statistically using one-way ANOVA followed by Dunnett's multiple comparisons test ($n = 3$ in (**A**, **B**); $n = 6$ for 0.5 mg/kg group and $n = 3$ for the others in **C**–**E**). Asterisks indicate a significant difference (*$P < 0.05$, **$P < 0.005$, ***$P < 0.0005$, ****$P < 0.0001$). The difference between different dose groups at the same timepoint was analyzed statistically using two-way ANOVA followed by Sidak's multiple comparisons test. Sign cross characters indicate a significant difference ($^{++++}P < 0.0001$, $^{+++}P < 0.0005$, $^{++}P < 0.005$, $^{+}P < 0.05$). This experiment was repeated once with similar result obtained.

therapy approaches have made significant progress in attempts to achieve a sustainable therapeutic effect, notably, the ex vivo gene therapy using autologous hematopoietic stem cells (HSC) modified with lentiviral vector and in vivo gene therapy using adeno-associated virus (AAV). Products of these two modalities are in clinical trials aiming for long-term genetic correction by a single treatment [30, 31]. Moreover, AAV or HSC may potentially be able to cross the blood-brain barrier therefore to benefit those GD patients with neurological manifestations [32–34]. However, challenges remain formidable precluding wide application of such gene therapies in GD treatment. AAV and ex vivo HSC are complex biological products that are difficult to develop and manufacture so that are likely to incur a high price [35]. Additionally, there are still concerns about unintended genomic insertion in host cells when using lentiviral and AAV vectors with the potential to develop malignancies, albeit at a low rate [36]. By contrast, an LNP-mRNA drug mitigates the safety concerns and manufacturing issues associated with cell culture and production of viral vectors. Efficient target protein expression can be achieved intracellularly with LNP delivered mRNA without the risk of genetic integration. Furthermore, the LNP-mRNA therapy is suitable for repeat dosing, and individual dose adjustment is feasible for long-term disease management owing to the transient nature of mRNA molecules.

In this study, JCXH-301/JCXH-301m, LNP-encapsulated mRNA encoding *GBA1*, was evaluated as a novel drug for GD treatment. Our results showed that JCXH-301m was distributed in various tissues after i.v. administration in mice. Intracellular β-GCase expression was found predominantly in macrophages and DC in the spleen and bone marrow. In a GD mouse model, JCXH-301m therapy restored the expression of enzymatically active β-GCase, resulting in significant reduction of lyso-GL1. These effects were observed in peripheral blood as well as in liver and spleen, the main organs affected in human GD patients. Noticeably, a trend of elevated *GBA1* mRNA level was also found in the brain, suggesting that i.v. administered JCXH-301 might be able to compensate the β-GCase deficiency in the central nerve system therefore to benefit the neurologic types of GD. However, it is yet to investigate whether the increase is caused by JCXH-301 transfected endothelial cells in the brain or circulating monocytes.

When compared to Cerezyme, which has a half-life no more than 12 h in the serum, liver and spleen, JCXH-301/JCXH-301m manifested a clear benefit in achieving considerably longer β-GCase expression after i.v. injection, persisting up to 3–7 days in the serum and 7–14 days in the liver and spleen. Since LNP-mRNA delivers the *GBA1* gene intracellularly, JCXH-301 has another advantage in that it compensates the β-GCase deficiency directly in the cells, thereby more efficiently reduces the cytoplasmic accumulation of toxic substrates, particularly in such organs of reticuloendothelial system as liver, spleen and bone marrow. JCXH-301m was shown to be well tolerable after repeat-dosing with a transient and minimal innate cytokine production being detected as compared to positive control of Con A treatment.

In summary, we describe a novel LNP-mRNA-based prototype drug that restores the β-GCase deficiency in D427V Gaucher mice. JCXH-301 has potential to become an alternative to current protein-based ERT with advantages in its broad tissue distribution, prolonged pharmacodynamic duration and direct intracellular restoration of β-GCase deficiency by mRNA. Although sustained substrate-reduction can be achieved with in vivo AAV or ex vivo HSC gene therapies based on preclinical evidence [37, 38], the transient nature of mRNA has an advantage in mitigating any safety concern associated to genetic integration by viral vectors. Using the dose conversion method described by Nair et al. [39], a JCXH-301 dose of 0.5 mg/kg in mice is equivalent to 0.04 mg/kg in human. It is encouraging to note that satisfactory tolerability has recently been shown in phase 1/2 clinical trials for LNP delivered RNA therapies using doses 10 to 20 times above this level [40, 41]. If JCXH-301 is successfully developed, a marketed price below a hundred US dollars ($) per dose can be achievable for 0.04 mg/kg a human dose by our estimation. Hence, the estimated expense for JCXH-301 would be about $2600 per annum or ~$26,000 a decade on a treatment regimen of once every two weeks. This would represent a dramatic cost reduction as compared to the current ERT of $150,000 to $700,000 per year on average [42] and to emerging in vivo AAV or ex vivo HSC gene therapy approaches of $2–3 million per treatment [35], offering a more sustainable option for lifetime management of GD patients. These results support the further development of JCXH-301 for the treatment of GD patients.

DATA AVAILABILITY

The sequences generated in this study are available in the NCBI database (Accession Gene ID. 2629 and 14466). The other data related to this study are available from the corresponding author upon request, with the permission of Immorna Biotechnology/ Immorna Biotherapeutics.

REFERENCES

1. Sheth J, Bhavsar R, Mistri M, Pancholi D, Bavdekar A, Dalal A, et al. Gaucher disease: single gene molecular characterization of one-hundred Indian patients reveals novel variants and the most prevalent mutation. BMC Med Genet. 2019;20:31.
2. Stirnemann J, Belmatoug N, Camou F, Serratrice C, Froissart R, Caillaud C, et al. A review of gaucher disease pathophysiology, clinical presentation and treatments. Int J Mol Sci. 2017;18:441.
3. Vardi A, Zigdon H, Meshcheriakova A, Klein AD, Yaacobi C, Eilam R, et al. Delineating pathological pathways in a chemically induced mouse model of Gaucher disease. J Pathol. 2016;239:496–509.
4. Hruska KS, LaMarca ME, Scott CR, Sidransky E. Gaucher disease: mutation and polymorphism spectrum in the glucocerebrosidase gene (GBA). Hum Mutat. 2008;29:567–83.
5. Cox TM, Cachon-Gonzalez MB. The cellular pathology of lysosomal diseases. J Pathol. 2012;226:241–54.
6. Panicker LM, Miller D, Awad O, Bose V, Lun Y, Park TS, et al. Gaucher iPSC-derived macrophages produce elevated levels of inflammatory mediators and serve as a new platform for therapeutic development. Stem Cells. 2014;32:2338–49.
7. Kishnani PS, Al-Hertani W, Balwani M, Goker-Alpan O, Lau HA, Wasserstein M, et al. Screening, patient identification, evaluation, and treatment in patients with Gaucher disease: Results from a Delphi consensus. Mol Genet Metab. 2022;135:154–62.
8. Weinreb NJ. Imiglucerase and its use for the treatment of Gaucher's disease. Expert Opin Pharmacother. 2008;9:1987–2000.
9. Murugesan V, Chuang W-L, Liu J, Lischuk A, Kacena K, Lin H, et al. Glucosylsphingosine is a key biomarker of Gaucher disease. Am J Hematol. 2016;91:1082–9.
10. Gary SE, Ryan E, Steward AM, Sidransky E. Recent advances in the diagnosis and management of Gaucher disease. Expert Rev Endocrinol Metab. 2018;13:107–18.
11. Yap TL, Velayati A, Sidransky E, Lee JC. Membrane-bound alpha-synuclein interacts with glucocerebrosidase and inhibits enzyme activity. Mol Genet Metab. 2013;108:56–64.
12. Badieyan ZS, Evans T. Concise review: application of chemically modified mRNA in cell fate conversion and tissue engineering. Stem Cells Transl Med. 2019;8:833–43.
13. Nelson J, Sorensen EW, Mintri S, Rabideau AE, Zheng W, Besin G, et al. Impact of mRNA chemistry and manufacturing process on innate immune activation. Sci Adv. 2020;6:eaaz6893.
14. Vaidyanathan S, Azizian KT, Haque A, Henderson JM, Hendel A, Shore S, et al. Uridine depletion and chemical modification increase Cas9 mRNA activity and reduce immunogenicity without HPLC purification. Mol Ther Nucleic Acids. 2018;12:530–42.
15. Xu S, Yang K, Li R, Zhang L. mRNA vaccine era-mechanisms, drug platform and clinical prospection. Int J Mol Sci. 2020;21:6582.
16. Martini PGV, Guey LT. A new era for rare genetic diseases: messenger RNA therapy. Hum Gene Ther. 2019;30:1180–9.
17. Bernal JA. RNA-based tools for nuclear reprogramming and lineage-conversion: towards clinical applications. J Cardiovasc Transl Res. 2013;6:956–68.
18. Berraondo P, Martini PGV, Avila MA, Fontanellas A. Messenger RNA therapy for rare genetic metabolic diseases. Gut. 2019;68:1323–30.

19. An D, Frassetto A, Jacquinet E, Eybye M, Milano J, DeAntonis C, et al. Long-term efficacy and safety of mRNA therapy in two murine models of methylmalonic acidemia. EBioMedicine. 2019;45:519–28.
20. Dekker N, van Dussen L, Hollak CEM, Overkleeft H, Scheij S, Ghauharali K, et al. Elevated plasma glucosylsphingosine in Gaucher disease: relation to phenotype, storage cell markers, and therapeutic response. Blood. 2011;118:e118–e27.
21. Polinski NK, Martinez TN, Gorodinsky A, Gareus R, Sasner M, Herberth M, et al. Decreased glucocerebrosidase activity and substrate accumulation of glycosphingolipids in a novel GBA1 D409V knock-in mouse model. PLOS ONE. 2021;16:e0252325.
22. Sardi SP, Clarke J, Kinnecom C, Tamsett TJ, Li L, Stanek LM, et al. CNS expression of glucocerebrosidase corrects -synuclein pathology and memory in a mouse model of Gaucher-related synucleinopathy. Proc Natl Acad Sci USA. 2011;108:12101–6.
23. Xu YH, Quinn B, Witte D, Grabowski GA. Viable mouse models of acid beta-glucosidase deficiency: the defect in Gaucher disease. Am J Pathol. 2003;163:2093–101.
24. Xu Y-H, Sun Y, Barnes S, Grabowski GA, Schiffmann R. Comparative therapeutic effects of velaglucerase alfa and imiglucerase in a Gaucher disease mouse model. PloS ONE. 2010;5:e10750.
25. Zheng W, Du S, Tian M, Xu W, Tian Y, Li T, et al. Lepidium meyenii Walp exhibits anti-inflammatory activity against ConA-induced acute hepatitis. Mediat Inflamm. 2018;2018:8982756.
26. Ji YR, Kim HJ, Bae KB, Lee S, Kim MO, Ryoo ZY. Hepatic serum amyloid A1 aggravates T cell-mediated hepatitis by inducing chemokines via Toll-like receptor 2 in mice. J Biol Chem. 2015;290:12804–11.
27. de Fost M, Aerts JM, Hollak CE. Gaucher disease: from fundamental research to effective therapeutic interventions. Neth J Med. 2003;61:3–8.
28. Campbell TN, Choy FY. Knockdown of chimeric glucocerebrosidase by green fluorescent protein-directed small interfering RNA. Genet Mol Res. 2004;3:282–7.
29. Cox TM. Gaucher disease: clinical profile and therapeutic developments. Biologics. 2010;4:299–313.
30. Kido J, Sugawara K, Nakamura K. Gene therapy for lysosomal storage diseases: current clinical trial prospects. Front Genet. 2023;14:1064924.
31. Hughes DA, Ferrante F. GALILEO-1: a phase I/II safety and efficacy study of FLT201 gene therapy for Gaucher Disease Type 1. Future Rare Dis. 2023;3:FRD35.
32. Rastall DP, Amalfitano A. Recent advances in gene therapy for lysosomal storage disorders. Appl Clin Genet. 2015;8:157–69.
33. Sevin C, Deiva K. Clinical trials for gene therapy in lysosomal diseases with CNS involvement. Front Mol Biosci. 2021;8:624988.
34. Du S, Ou H, Cui R, Jiang N, Zhang M, Li X, et al. Delivery of glucosylceramidase beta gene using AAV9 vector therapy as a treatment strategy in mouse models of gaucher disease. Hum Gene Ther. 2019;30:155–67.
35. Tucci F, Scaramuzza S, Aiuti A, Mortellaro A. Update on clinical ex vivo hematopoietic stem cell gene therapy for inherited monogenic diseases. Mol Ther. 2021;29:489–504.
36. Hudry E, Vandenberghe LH. Therapeutic AAV gene transfer to the nervous system: a clinical reality. Neuron. 2019;102:263.
37. Dahl M, Smith EMK, Warsi S, Rothe M, Ferraz MJ, Aerts J, et al. Correction of pathology in mice displaying Gaucher disease type 1 by a clinically-applicable lentiviral vector. Mol Ther Methods Clin Dev. 2021;20:312–23.
38. Corbau R, Miranda C, Comper F, Kalcheva P, Chisari E, Cocita C, et al. FLT201: an AAV-mediated gene therapy for type 1 Gaucher disease designed to target difficult to reach tissues. Mol Genet Metab. 2021;132:S28–S9.
39. Nair AB, Jacob S. A simple practice guide for dose conversion between animals and human. J Basic Clin Pharm. 2016;7:27–31.
40. Koeberl D, Schulze A, Sondheimer N, Lipshutz GS, Geberhiwot T, Li L, et al. Interim analyses of a first-in-human phase 1/2 mRNA trial for propionic acidaemia. Nature. 2024;628:872–7.
41. Cohn DM, Gurugama P, Magerl M, Katelaris CH, Launay D, Bouillet L, et al. CRISPR-based therapy for hereditary angioedema. N. Engl J Med. 2025;392:458–67.
42. Pharmacoeconomic Review Report: Eliglustat (Cerdelga): (Sanofi Genzyme): Indication: Gaucher Disease Type 1 [Internet]. CADTH Common Drug Reviews. Ottawa (ON): Canadian Agency for Drugs and Technologies in Health; 2017.

ACKNOWLEDGEMENTS

We thank Shanghai Model Organisms Center Inc. and Ms. Dongli Liang of Shanghai Jiaotong University for their expert assistance in mouse experiments. We thank Mr. Ji Xu and Ms. Yanqi Su for the logistical support.

AUTHOR CONTRIBUTIONS

Z.W., Z.G., S.W., Y.C., and Y.L. developed the concept and designed the studies. S.W. and Z.Z performed the experiments and curated the data. S.W., Z.Z., and Y.C. analyzed the data. S.W., Y.C., X.Q., and Y.L. wrote the manuscript. All the authors reviewed the manuscript and agreed this version to be published.

FUNDING

The study was fully funded by Immorna Biotechnology/Immorna Biotherapeutics.

COMPETING INTERESTS

All authors are employees of Immorna Biotechnology/Immorna Biotherapeutics. Z.W., Y. L., and Z. G. are inventors to the Chinese patent CN118903476B.

ETHICS APPROVAL STATEMENT

All animal experiments were performed in accordance with the recommendations in the Guide for the Care and Use of Laboratory Animals and relevant Chinese laws and regulations. All animal use adhered to the IACUC guidelines for animal ethics and welfare. The in vivo experiment protocols were approved by the Institutional Animal Care and Use Committee (IACUC) of Shanghai Jiaotong University, China (authorization number: A2022044), and Shanghai Model Organisms Center, Inc., China (authorization number: No. 2020-0047).

ADDITIONAL INFORMATION

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41434-025-00549-2.

Correspondence and requests for materials should be addressed to Yuanqing Liu.

Reprints and permission information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g., a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.